

10 Gigabit Ethernet

Module 26

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Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar University of Science and Technology Hisar harana I welcome you all to the video lecture series of cyber and information security in this lecture we will study about 10g ethernet which is basically used for data transmission on wired Lan at speed of 10 gbps first of all please see the contents which I shall cover after covering brief introduction about 10 gbit ethernet I shall take up layered protocol architecture of 10 GB ethernet which will be followed by physical and Mac layer of 10 GB ethernet in last I shall take up 10 GB ethernet standards so now let us start as the demand for highspeed Network grow the need for a faster ethernet technology is

apparent in March 1999 a working group was formed as higher speed study group hssg to develop a standard for 10 gab ethernet 10 gab ethernet being standardized as i e 823a is a telecommunication technology that offers data speeds up to 10 billion bits per second 10 gabit ethernet called as 10 G or 10 GBE or 10 gig e is a group of computer networking Technologies for transmitting ethernet frames at a rate of 10 gab bits per second it was first defined by the i e 802.3 AE 2002 standard unlike previous ethernet standard standards 10 GB ethernet defines only full duplex Point to point links which are generally connected by Network switches in gigabit Ethernet shared medium csma/cd operation has not been carried over from the previous

generation ethernet standards half duplex operation and hubs do not exist in 10 GBE like previous versions of ethernet 10 GBE can use either copper or Fiber cabling however because of its bandwidth requirement higher grade copper cables are required like Category 6 6A or category 7 cables for links up to 100 m different I E 802.3 standard for 10g ethernet are as shown in table i e 802.3 AE reified in 2002 supports 10 gbits per second over fiber for landan and when i e 802.3 a K defined in 2004 supports 10 gbit per second over twin XL cable i e 802.3 2005 is a revision of Base standard incorporating the prior amendments and era i e 802.3 n defined in 2006 supports 10 gbits per second over copper twisted pair cable i e

802.3 a defined in 2007 is back plane ethernet which supports 1 and 10 gbits per second over printed circuit boards i e 802.3 AQ defined in 2006 supports 10 gbits per second over multimode fiber with enhanced Equalization i e 802.3 2008 is a revision of Base standard i e 82.3 AV defined in 2009 is 10 gbit per second ethernet physical for Eon i e 82.3 2012 is latest version of Base standard now let us see layered protocol architecture of 10 GB ethernet the ethernet protocol basically implements the bottom two layers of the open system interconnect OSI model that is the data link and physical sublayers figure depicts the typical ethernet protocol stack and the relationship to the OSI model physical

layer of OSI is mapped onto PMD PMA PCS and reconciliation Su layer RS Network layer of OSI is mapped onto map and LLC layer basic mediums supported by 10g ethernet R 10g base R for Len fiber 10g Bas W for Van fiber and 10 G base x for copper the architecture of ethernet subdivides the physical layer into three sub layers the physical medium dependent PMD sublayer the physical medium attachment PMA and a physical coding sublayer PCS PMD physical medium dependent the PMD sublayer is responsible for signaling transmission the typical PMD functionality includes amplifier modulation and wave shaping different PMD devices May support different media PCS physical coding sublayer the PCS sublayer consist of coding and a

serializer or multiplexing functions the structure of this layer is defining feature that differentiates the Lan and Van physical for the van physical the PCS operates in a serialized fashion for the L physical it operates in two different modes first is parallel mode in which there is a multiplexer that multiplexes the data onto four 2.5 gbps lines second is serial L that acts as a serializer that serializes and deserializes the data to and from a single channel the serial Lan physical uses a 64bit by 66 bit coding scheme for the serial when physical and additional sublayer the van interface sublayer W is required between the serial PCS and the serial PMA the Wiis sublayer ensures operability with Sonet by including a

simplified Sonet frame in w p ma physical medium attachment the PMA sublayer is responsible for serialization of code groups into bit stream suitable for serial bit oriented physical devices and vice versa it supports multiple encoding schemes in support of the different pmds that use an encoding that is specific to the medium it supports reconciliation sublayer the reconciliation sublayer acts as a command translator it Maps the terminology and commands used in the Mac layer into electrical formats appropriate for the physical layer entities medium access control Mac the medium access control sub layer provides a logical connection between the Mac clients of itself and its peer station its main responsibility is to initialize

control and manage the connection with the peer station xgm I 10 gigabit media independent interface it provides a standard interface between the Mac layer and the physical layer it isolates the Mac layer and the physical layer enabling the Mac layer to be used with various implementations of the physical layer MDI medium dependent interface it defines different

connector types for different physical media and BD devices first let us see physical layer architecture of 10g ethernet one of the first things that anyone familiar with ethernet would notice about 10 gabit variant is that there are two options for implementing the physical layer first is serial solution and second is parallel solution the serial solution uses is one highspeed

10gbps PCS PMA PMD circuit block and the parallel solution uses multiple PCS PMA PMD slow speed links Each of which operates at lower speed in the serial implementation there is one physical Channel operating at 10 GBS per second as shown in figure the operation is straightforward for transmission the reconciliation module passes the signals corresponding to the make data word by word to the PCS module the PCS module then encodes the signals with a predefined coding technique and passes the encoded signal to the PMA module the PMA module then serializes the encoded signals and passes the stream to the PMD module the PMD module transmits the signal stream over the fiber at 10 gbits per second on receiving side the

process is reverse the main advantage of the serial architecture is that the transmitting receiving operation is straightforward it does not require a complicated multiplexing or demultiplexing that is needed in the parallel implementation in parallel implementation there are multiple physical channels say n subchannels that may be implemented by using parallel cables or wdm multiplexing for transmission the distributor multiplexes the data accepted from the Mac layer into n streams in the round robin motion each stream is given to each PCS module each PCS module encodes the received stream and passes it to each PMA module for serialization after serialization each PMD module transmits each serialized data stream at the rate

of 10 divided by n gbits per second on receiver side reverse process is done the process is shown in figure the main advantage of the parallel implementation is that the operating rate in the PCS PMA module is reduced which enables cheaper devices like cimos or bipolar to be used the disadvantages are the need of distributor or collector module that may be sensitive to timing Jitters and the usage of multiple sets of logic circuits and Laser equipment now let us see Mac layer of 10g ethernet the Mac layer of 10 GB ethernet is very similar to the Mac layer of previous ethernet Technologies it uses the same ethernet address and frame formats but it does not support the half duplex mode it supports data rates of less than

10 gbps using a pacing mechanism for rate adaptation and flow control it supports full duplex mode only in previous ethernet standards there were two modes of operation half duplex and full duplex the half duplex mode has been defined since the original version of ethernet in this mode data is transmitted using the popular CSM by CD protocol on a shared medium its Simplicity contributed to the early success of the ethernet standard this mode of operation is so famous that many wrongly associate the csma by CD protocol with standard ethernet operation efficiency and distance limitation are the main disadvantages of the half duplex mode in this mode the link distance is limited by the minimum Mac frame size this restriction

reduces the efficiency

drastically for high rate transmission at the transmission rate of 10 gbps the half duplex mode is not an attractive option and no Market would realistically exist for the half duplex operation at this rate of transmission as most of the Links at 10 gbps are point to point over Optical fibers in this case the full duplex operation would be the preferred option so the standard for 10 GB ethernet specifies only the full duplex operation in full duplex operation there is no contention the Mac layer entity can transmit whenever it wants provided that its Speer is ready to receive Mac frame format the key purpose of developing 10 GB ethernet standard is to use the same Mac frame format as specified in the preceding ethernet

standards this will allow a seamless integration of the 10 GB ethernet with the existing ethernet networks since the full duplex operation is used the link distance does not affect the Mac frame size the minimum Mac frame size will be made equal to 6 four octets as specified in the previous ethernet standards the macrame format is depicted in figure carrier extension is not required here the 10g ethernet frame format consist of the following Fields Preamble PR it consist of seven bytes the Preamble is an alter alternating pattern of ones and zeros that tells receiving stations that a frame is coming and that provides a means to synchronize the frame reception portion of receiving physical layers with the incoming bit stream start of frame delimiter s

oof it consist of one byte the sof is an alternating pattern of ones and Zer ending with two consecutive one bits indicating that the next bit is the leftmost bit in the leftmost bite of the destination address destination address da consist of six bytes the da field identifies which station should receive the frame the leftmost bit in the da field indicates whether the address is an individual address indicated by a zero or a group address indicated by a one the second bit from the left indicates whether the da is globally administered indicated by a zero or locally administered indicated by a one the remaining 4 six bits are a uniquely assigned value that identifies a single station a defined group of station or all stations on the

network Source address sa it consist of six bytes the sa field identifies the sending station the sa is always an individual address and the leftmost bit in the essay field is always zero length type consist of two bytes this field indicates either the number of Mac client databased bytes that are contained in the data field of the frame or the frame type ID if the frame is assembled using an optional format if the length type field value is less than or equal to 1500 the number of LLC bytes in the data field is equal to the length type field value data it is a sequence of n bytes of any value where n is less than or equal to 1,500 if the length of the data field is less than 46 the data field must be

extended by adding a filler sufficient to bring the data field length to 46 bytes frame check sequence FCS consist of 4 bytes this sequence contains a 32bit cyclic redundancy check CRC value which is created by the sending Mac and is recalculated by the receiving magc to check

for damaged frames the FCS is generated over the data length type and data fields so this was frame format of 10g ethernet now let us see standards of 10g ethernet the 10 GB ethernet standard includes a number of different physical layer standards they are broadly divided into three categories first category is Lan fiber which supports 10g Bas Sr 10 G Bas LR M 10 G Bas LR 10 G base E 10 G base lx4 and 10 G G base ZR type of physical standard second category is V fiber

which supports 10 G Bas s SW 10 G Bas LW and 10 G Bas ew type of physical standard third category is len copper which supports 10g base CX4 10g base p 10g base kx4 and 10g base KX type of standard first of all let us see Lan using fiber the most common Optical variety is referred to as Lan physical used for connecting directly between routers and switches it is basically indicated by alphabet R in ethernet standard although 25 minutes, 1 second called Lan this can be used with fiber cable up to 80 kilomet Lan physical uses a line rate of 10.3 gbits per second and a 66 bit encoding different standards used for fiber Lan are as shown in table 10g Bas Sr R it is short reach standard defined in 2002 and designed to support short distances over deployed

serial multimode fiber cabling with 850nm wavelength it delivers serialized data at a line rate of 10.3125 gbits per second its range is 300 to 400 26 minutes, 1 second m it falls under i e 802.3 AE standard 10g Bas LR it is a long range Optical technology defined in 2002 delivering serialized 10 GB ethernet over 1310 NM single mode fiber using 64bit by 66 bit physical coding sublayer PCS it delivers serialized data at a line rate of 10.3125 gabs per second its range is 10 kilom it falls under i e 802.3 AE standard 10g base ER it is is an extended range fiber defined in 2002 which supports distances up to 40 kilom over single mode fiber with 1550 NM wavelength and speed of 10.3125 gbits per second recently several

manufacturers have introduced 80 kilom range ER plugable interfaces it falls under i e 802.3 AE standard 10g base lx4 it was defined in 2002 using wavelength division multiplexing to support ranges of 300 M over deployed multimode cabling and 10 km over single mode fiber using wavelength of 1310 NM this is achieved through the use of four separate laser sources operating at 3.125 gabs per second on unique wavelengths it falls under i e 802.3 AE standard it is now getting replaced by 10g Bas lrm now 10g Bas lrm it is Long Reach multim mode standard defined in 2006 which support distances up to 22m on fddi grade 62.5 micromet multi mode cable with 1310 NM wavelength which was originally installed in early 1990s for 100 Mbps networks and it

offers speed of 10.3125 GBS per second it falls under i e 802.3 AQ standard next we shall see van using fiber 10g Bas s SW 10g base LW and 10g Bas ew are varieties that use the van physical the when physical variants correspond at the physical layer to 10g Bas Sr 10gbase LR and 10g Bas ER respectively and hence use the same type of fiber and support the same distances it falls under i e 802.3 AE standard now moving on to Copper Lan 10g base CX4 this standard was designed in 2002 and is also known by its working group name i e 802.3 a k it

transmits over four lanes in each Direction Over copper CA using infiniband 4X twin XL 8 pair it is designed to work up to a distance of 15 M only this technology has the lowest

cost per Port among all 10 GB interconnects at the expense of range each lane of the copper carries 3.125 GHz of signaling bandwidth 10g base T it is a newly released standard by the IEEE 802.3 committee defined in 2006 to provide 10 gabit per second connections over conventional unshielded or shielded twisted pair cables of category 6 6A and 7 category 6 supports 55 M distance and Category 6 a and 7 supports 100 m distance this standard is defined as IEEE 802.3 an 10g base kx4 and 10g Bas KR it is a back plane ethernet also known know by its task force name IEEE 802.3 a and is used in Back plane applications such as blade servers and modular router or switches with upgradable line cards 802.3 AP implementations are required to operate in an environment comprising up

to 1 M of proper printed circuit board with two connectors the standard defines two Port types for 10 gabit per second that is 10g base kx4 and 10 gbas KR new back plane designs use 10g base KR which uses single back plane Lane rather than 10g Bas KX x 4 which operates over four back plane Lanes apart from all the standards explained till now there is one more standard called 10g base PR 10g Bas PR that is P40 originally specified in IEEE 802.3 AV is a 10g ethernet physical for Passive Optical networks and uses 157 nanometer lasers in the downstream Direction and 1270 nanometer lasers in the Upstream Direction on Downstream side it delivers serialized data at a line rate of 10.3125 gabs per second in a point to multi-point configuration

10g Bas PR has three power budgets specified as 10g Bas pr1 10 gbas pr20 and 10 gbas pr30 so friends this was 10 GB ethernet finally summarizing we learned about introduction to 10 GB ethernet which was followed by layer protocol architecture and details of physical and Mac layer of 10g ethernet in the end we discussed different standards of 10 gbit ethernet hope the concepts explained in this lecture were understandable and helpful see you in the next lecture till then goodbye enjoy the day thank you you